DEEP LEARNING • COURSE PRESENTATION

The Evolution of Artificial Intelligence

From Rule-Based Logic to Generative Intelligence — Tracing Six Decades of Machine Reasoning

NAME	Dakshraj Singh Chandawat
ROLL NO.	2411021240050
COURSE	BCA – Artificial Intelligence & Machine Learning
SEMESTER	5th Semester

INTRODUCTION

Understanding Artificial Intelligence

Artificial Intelligence refers to the branch of computer science that enables machines to perform tasks that normally require human intelligence, such as reasoning, learning from experience, recognising patterns, and making decisions. The journey of AI has not been one smooth advance but a series of distinct waves, each shaped by the computing power, data, and algorithms available at the time. Understanding this evolution explains why today's systems can write essays and generate images with such fluency.

- Artificial Intelligence blends computer science, mathematics, and cognitive psychology to build systems that reason and learn rather than only follow fixed instructions.
- The field has passed through repeated cycles of excitement and disappointment, known as AI summers and AI winters, driven by funding and unmet expectations.
- Three factors converged to accelerate modern AI: massive internet-scale datasets, affordable GPU computing power, and refined deep learning algorithms.
- Key takeaway: AI's progress is best understood as an evolving relationship between data, hardware, and algorithms, not a single invention.

$184B+

Global AI market value, 2024 (Statista)

FIVE ERAS OF AI

1950s–80s
Early AI
Rule-based logic

1990s–2000s
Machine Learning
Statistical patterns

2010s
Deep Learning
Neural networks

2020s
Generative AI
Transformers & LLMs

2025+
Future AI
AGI & responsible AI

ERA 1 OF 4

Early AI: The Age of Rules and Logic

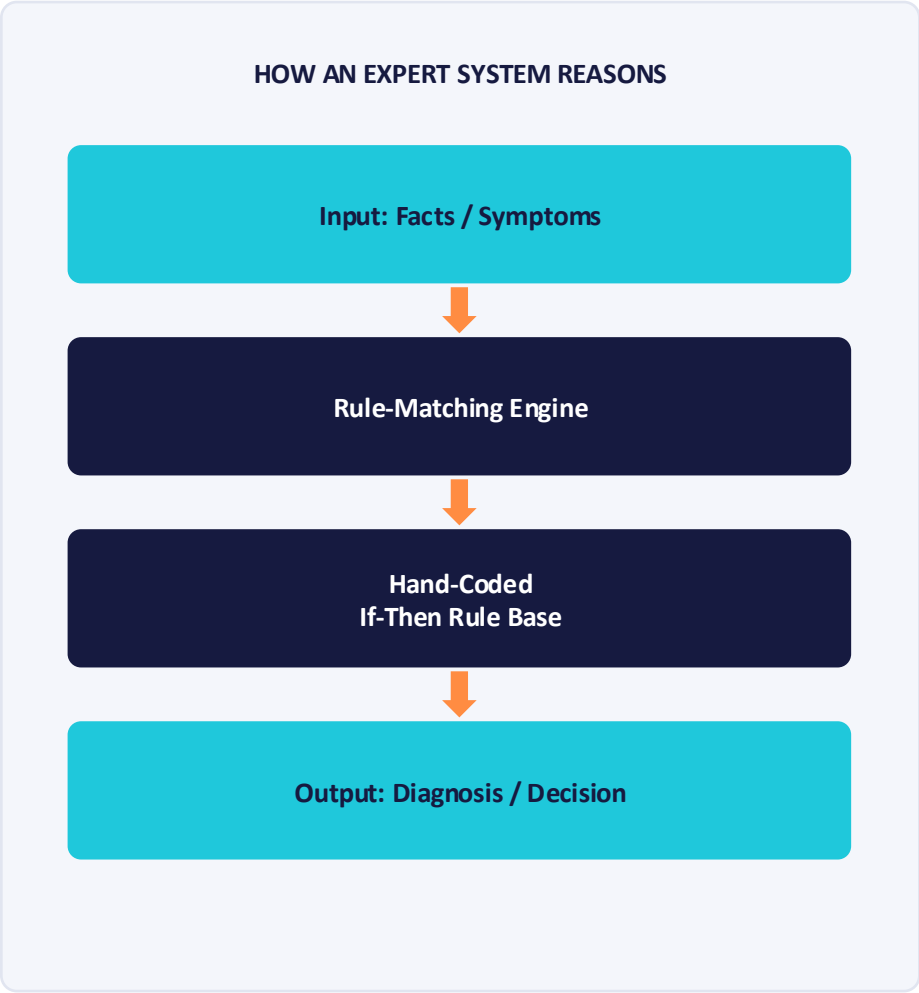
1950s – 1980s

The story of AI formally begins in 1956 at the Dartmouth Conference, where researchers proposed that every feature of learning could, in principle, be described precisely enough for a machine to simulate it. Early AI systems, known as symbolic or rule-based AI, did not learn from data at all. Programmers instead hand-coded explicit logical rules, so a machine could only ever be as capable as the knowledge it was directly given.

- Alan Turing's 1950 paper proposed the Turing Test, asking whether a machine's conversation could be indistinguishable from a human's.
- Expert systems such as MYCIN and DENDRAL used thousands of if-then rules written by specialists to diagnose diseases or identify chemical compounds.
- These systems worked well in narrow, well-defined domains but failed when faced with ambiguity or situations outside their programmed rules.
- Limited computing power and the high cost of manually encoding knowledge led to funding cuts, producing the first AI Winter in the 1970s.

1956

Year the term "Artificial Intelligence" was coined at Dartmouth College



ERA 2 OF 4

Machine Learning: Teaching Computers to Learn from Data

1990s – 2000s

Instead of hand-coding every rule, machine learning let algorithms discover patterns directly from examples. A programmer would supply data and a general learning algorithm, and the system would adjust its internal parameters until predictions matched reality closely enough. This shift from explicit instructions to statistical pattern recognition marked one of the most important turning points in the history of computing.

- Algorithms such as decision trees, support vector machines, and Naive Bayes classifiers became popular for spam filtering and credit scoring.
- IBM's Deep Blue defeated world chess champion Garry Kasparov in 1997, proving search algorithms could outperform human intuition in narrow tasks.
- Machine learning still needs carefully engineered features chosen by human experts, unlike deep learning, which later learned to extract features automatically.
- Netflix launched its recommendation engine in the early 2000s using collaborative filtering to suggest films from patterns in millions of ratings.

TRADITIONAL PROGRAMMING vs MACHINE LEARNING

Aspect	Traditional	Machine Learning
Logic source	Written by programmer	Learned from data
Input	Rules + Data	Data + Answers
Output	Fixed results	Predictive model
Adapts to new data?	No	Yes
Example	Tax calculator	Spam filter

1997

IBM Deep Blue becomes the first computer to beat a reigning world chess champion

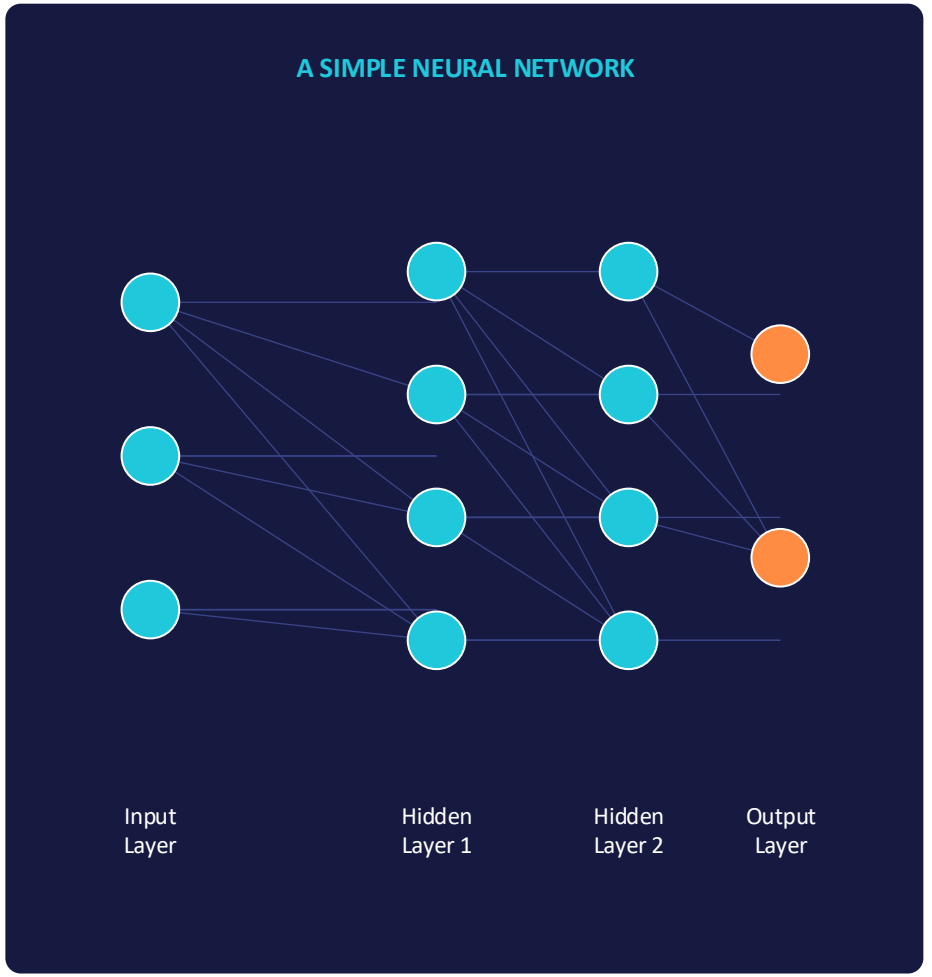
ERA 3 OF 4

Deep Learning: Neural Networks Take Over

2010s

Deep learning uses artificial neural networks with many stacked layers to automatically learn hierarchical representations of raw data, removing the need for manual feature engineering. Although the mathematics of neural networks dates back decades, deep learning only became practical once GPUs made it possible to train networks with millions of parameters on massive labelled datasets such as ImageNet.

- In 2012, a deep convolutional network called AlexNet won the ImageNet competition, cutting the error rate almost in half compared to rivals.
- NVIDIA's GPUs, originally built for gaming graphics, proved ideal for the parallel matrix computations that neural network training requires.
- Deep networks learn features automatically layer by layer, detecting edges in early layers and complex objects like faces in deeper layers.
- Google DeepMind's AlphaGo used deep reinforcement learning to defeat world champion Lee Sedol at Go in 2016, a game once thought too complex for machines.



15.3%

AlexNet's top-5 error rate in 2012, versus 26.2% for the next-best competitor

ERA 4 OF 4

Generative AI: Machines That Create

2020s

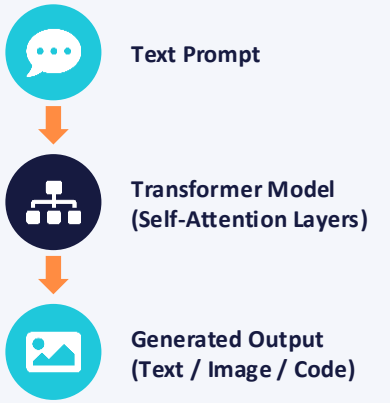
Generative AI is the newest wave, where deep learning models do not just classify or predict but create original text, images, audio, and code. Powered by a network design called the Transformer, introduced by Google researchers in 2017, these models are trained on enormous portions of internet text and can generate remarkably human-like, context-aware content on demand.

- OpenAI's ChatGPT, built on the GPT family of large language models, reached 100 million users within about two months of its late-2022 launch.
- The Transformer architecture uses self-attention, letting the model weigh the importance of every word in relation to every other word in a sentence.
- Image generators such as OpenAI's DALL-E and Google's Imagen turn a simple text description into a detailed, original picture within seconds.
- Microsoft integrated GPT-based models into Copilot across Office and GitHub, letting the AI draft documents, emails, and working program code.

100M

ChatGPT users reached within roughly two months of its public launch

HOW A GENERATIVE MODEL WORKS



BUILT ON GENERATIVE AI TODAY

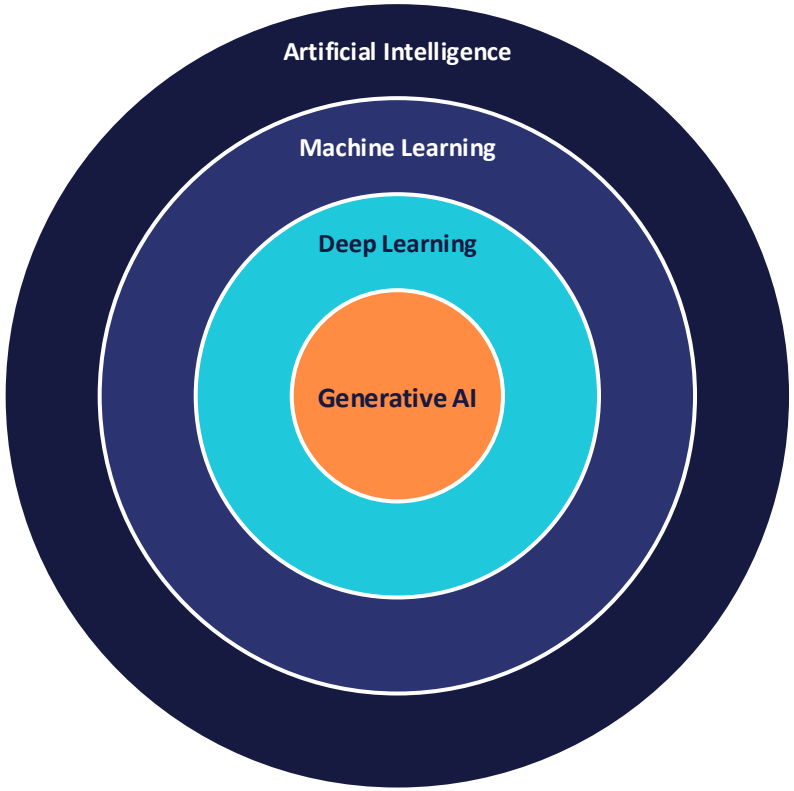


OpenAI ChatGPT | Google Gemini | Microsoft Copilot

CONCEPT MAP

AI vs Machine Learning vs Deep Learning vs Generative AI

These four terms are often used interchangeably, but they describe nested and increasingly specific ideas. Artificial Intelligence is the broad goal of building intelligent machines, Machine Learning is one approach that learns from data, Deep Learning is a specialised technique using layered neural networks, and Generative AI is a modern application of deep learning focused on creating new content.



- Artificial Intelligence is the umbrella field covering any technique, rule-based or data-driven, that gives machines human-like reasoning ability.
- Machine Learning is a subset of AI where systems improve automatically by learning statistical patterns from historical data instead of fixed rules.
- Deep Learning is a subset of ML that uses multi-layered neural networks to learn complex patterns directly from raw, unstructured data.
- Generative AI is a subset of deep learning built specifically to produce new original content such as text, images, music, or code.
- Key takeaway: every generative AI system is deep learning, machine learning, and AI — but the reverse is not true.

INDUSTRY IMPACT

AI Across Industries: Real-World Applications

The evolution from rule-based logic to generative intelligence has moved AI out of research labs and into everyday products used by billions of people. Leading technology companies now embed multiple generations of AI, sometimes classical machine learning, sometimes deep learning, within a single product, choosing whichever method best solves their specific business problem.

- Amazon uses machine learning across its recommendation engine, warehouse robotics, and Alexa voice assistant to personalise shopping and automate logistics.
- Tesla relies on deep learning-based computer vision, processing live camera feeds, to power its Autopilot and Full Self-Driving systems.
- Netflix applies collaborative filtering and deep learning to analyse viewing behaviour, driving an estimated 80 percent of content users watch.
- IBM Watson combined natural language processing and machine learning to defeat human champions on the quiz show Jeopardy in 2011.

AI IN ACTION

Amazon
Recommendations & logistics

Tesla
Autonomous driving vision

Netflix
Personalised content

IBM Watson
Natural language Q&A

Google
Search & multimodal AI

80%

Share of Netflix viewing activity estimated to be driven by its AI recommendation engine

LOOKING AHEAD

What Comes Next: Emerging Trends and Challenges

As AI systems grow more capable, research attention is shifting from raw performance toward reliability, transparency, and responsible deployment. The next phase of AI evolution is expected to focus on systems that reason more like humans, explain their own decisions, and operate safely within legal and ethical boundaries, rather than simply scaling up existing models further.

Artificial General Intelligence

A system with human-level reasoning across any task, pursued by labs like OpenAI and DeepMind.

Explainable AI

Making deep learning decisions transparent and interpretable, essential for healthcare, finance, and law.

Multimodal AI

Models such as Google's Gemini that reason across text, images, audio, and video together.

AI Regulation

Frameworks such as the EU AI Act introducing legal accountability for how AI is built and deployed.

ROAD AHEAD (PROJECTED)

- **2025** — Multimodal AI goes mainstream in consumer apps
- **2027** — Stronger global AI regulation takes effect
- **2030+** — Continued pursuit of AGI-level reasoning

EU AI Act

First comprehensive AI law passed by a major regulatory body, adopted 2024

CONCLUSION

Six Decades, One Continuous Story

Artificial Intelligence has evolved from hand-written logical rules, through statistical machine learning and deep neural networks, to generative systems capable of original creation. Each era did not discard the last but built directly on it, and the same pattern, more data, more compute, smarter algorithms, is likely to define whatever comes after generative AI.

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